



Responsible ML Audit of HMDA Loan Approval Model

GROUP 3



MEET THE TEAM - GROUP 3

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RESPONSIBLE MACHINE LEARNING CAPSTONE PROJECT

AGENDA - PHASE 1-2

- Overview
- Problem Statement & Motivation
- Dataset Overview
- Data Processing & Feature Engineering
- Modeling Approach Model Performance Comparison
- Model Explainability (SHAP Analysis)
- Failure Modes & Bias Risks





Problem Statement & Motivation

01

- Predict whether a loan application is approved or denied
- Dataset: HMDA mortgage lending data

02

- High-stakes decision → impacts access to credit
- Evaluate fairness - Risk of discrimination across racial groups
- Ensure compliance with fair lending laws

03

Goal: evaluate performance, fairness, and robustness

Dataset Overview

Content:

- Source: HMDA (Home Mortgage Disclosure Act) data
- Total rows in full dataset: 12,259,156
- Used stratified sample - 866K+ rows)
- Processed using DuckDB for scalability

KEY FEATURES



INCOME



LOAN AMOUNT



PROPERTY
VALUE

Data Processing & Feature Engine

- Removed invalid labels
- Created binary target
- Imputed missing values
- Removed biased features (interest_rate)
- No protected attributes used

14 - Final Features



Optimization Objective

What are we predicting and how do we measure success?

Binary

Classification Task

Macro F1

Primary Metric

$AIR \geq 0.80$

Fairness Constraint

3.12x

Class Imbalance Ratio

Label Creation

Action Taken 1 or 2 → **Label = 1 (Approved)**

Action Taken 3 →

Label = 0 (Denied)

Model Performance Comparison

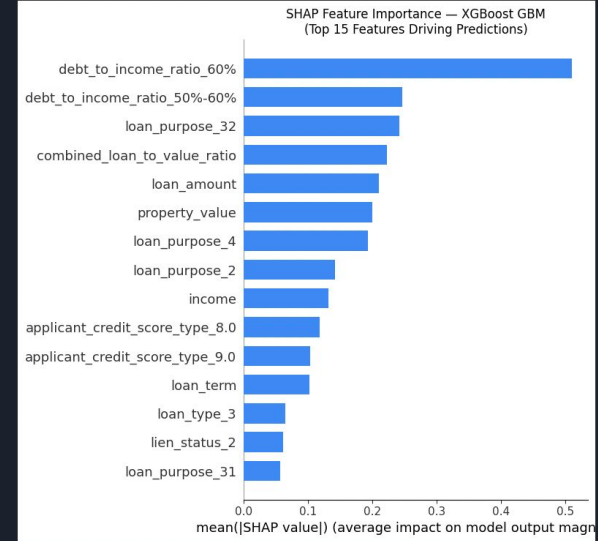
Model	Macro F1	ROC-AUC	F1 Denied
Random Forest (Baseline)	0.7926	0.8604	0.6711
XGBoost GBM @ 0.60 ✓	0.7233	0.8559	0.5015

XGBoost selected: industry standard for credit risk · superior SHAP explainability · threshold-optimized for fairness

SHAP Explainability

Top features with Proxy Bias:

- Loan amount
- Property value
- Income
- Debt-to-income ratio



Failure Modes

- Financial features act as proxies for race
- Model indirectly encodes bias
- Class imbalance: Overpredicts approvals
- Poor detection of denied loans (low recall)

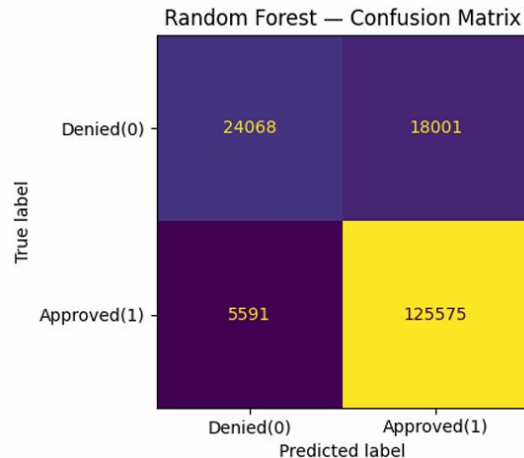
MODEL PERFORMANCE

Model	Macro F1	Balanced Acc	ROC-AUC
Random Forest	0.79	0.76	0.860
XGBoost	0.70	0.67	0.856

Key takeaway:

- Random Forest is more balanced
- XGBoost favors approvals

CONFUSION MATRIX



Key takeaway:

- Model struggles more with denied loans

Known Failure Modes

5 failure modes identified before running a single fairness metric

Class Imbalance

3.12x approved/denied ratio — model would ignore denials without correction

HIGH

Proxy Discrimination

Income & property_value are top SHAP drivers — can reconstruct race patterns

HIGH

HMDA Sentinel Values

Exempt, 1111, 9999 codes used instead of NaN — silent data corruption risk

MED

Missing Data by Class

interest_rate 100% missing for denied loans — would drop ALL denied records

HIGH

Historical Bias

HMDA reflects lender decisions, not creditworthiness ground truth

MED

Known Failure Modes

A good audit is not a perfect model. It is a justified model with documented risks.

Failure Mode	Mitigation Applied	Status
Class Imbalance	scale_pos_weight = 3.12 in XGBoost	RESOLVED
Proxy Discrimination	SHAP monitoring — flag if income/property_value shift rank	MONITORED
HMDA Sentinel Values	Recoded Exempt, 1111, 9999 → NaN before any cleaning	RESOLVED
Missing Data by Class	Removed interest_rate • Applied median imputation on remaining	RESOLVED
Historical Bias	AIR threshold monitoring + threshold adjustment mitigation	MONITORED
High FPR (50–70%)	Documented — reflects HMDA data structure, not model error	RESIDUAL

Residual risks are documented uncertainties — not incomplete work.
HMDA 2024 | Model Risk & Fair Lending Audit

AGENDA - PHASE 3

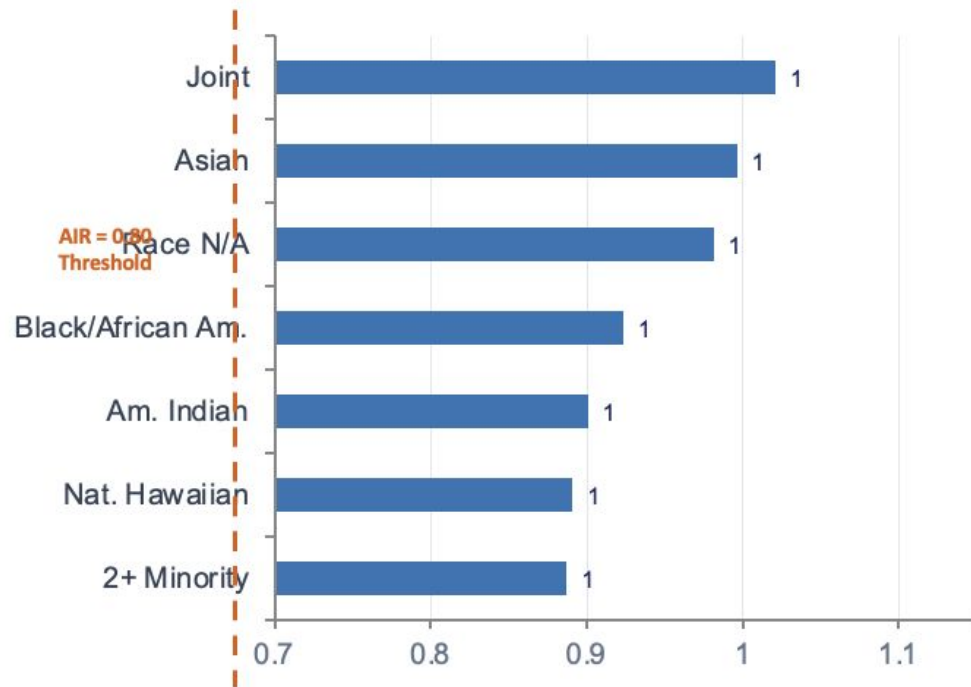
FAIRNESS

- Fairness Metrics Overview
- Adverse Impact Ratio (AIR) by Race
- False Negative Rate (FNR) Disparities
- Intersectional Fairness Analysis (Race $\times$ Sex)



Subgroup Errors — Adverse Impact Ratio (AIR)

Reference Group: White Applicants · AIR < 0.80 triggers fair lending scrutiny



Summary

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Groups Pass AIR ≥ 0.80

0.8872

Lowest AIR (2+ Minority)

0.9234

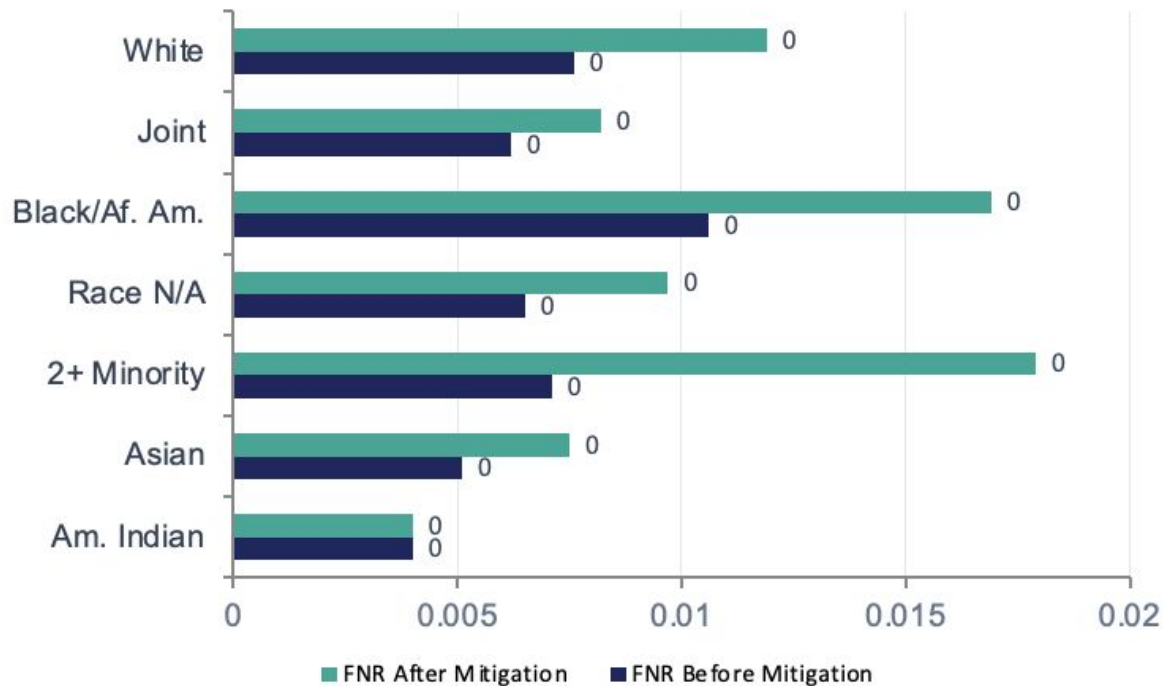
Black/African Am. AIR

0.60

Optimal threshold applied

Subgroup Errors — False Negative Rate (FNR)

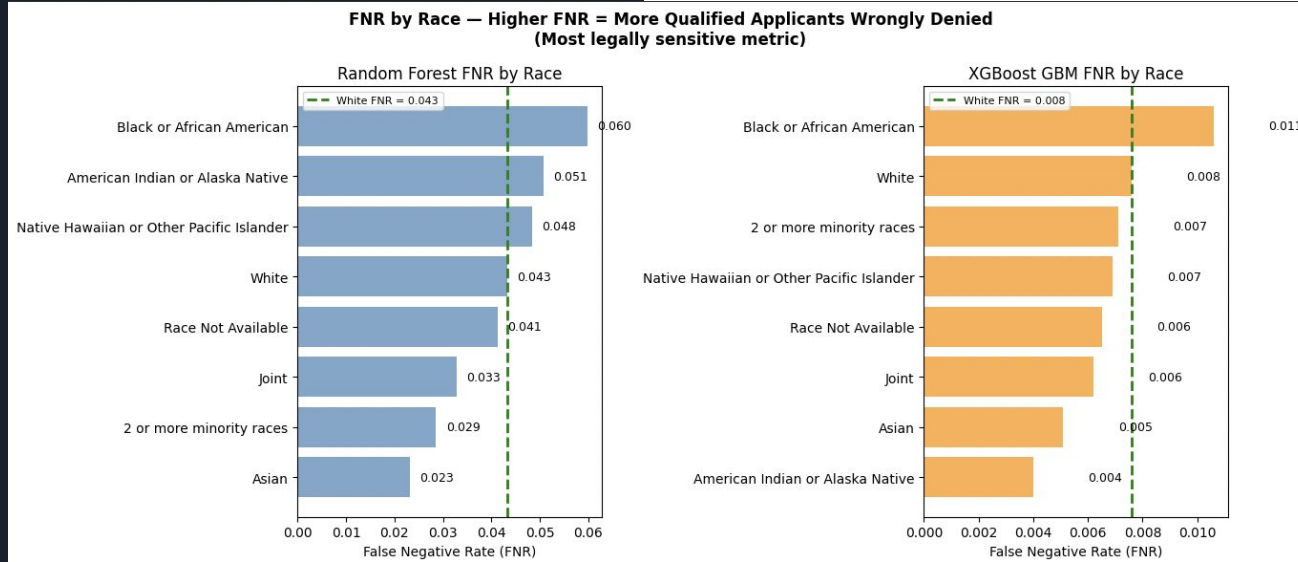
FNR = qualified applicants wrongly denied · Most legally sensitive metric under ECOA



Key Insights

- All FNR values are very low (0.4%–1.8%)
- Black/Af. Am. FNR rises slightly after threshold adjustment
- FPR is elevated (50–70%) — model over-approves across all groups
- HMDA data captures lender behavior, not creditworthiness ground truth

FNR Disparity



- Some groups have higher false negatives
- Meaning qualified applicants may be denied

Intersectional Analysis

- No major failures
- But small subgroup instability

AGENDA - PHASE 4

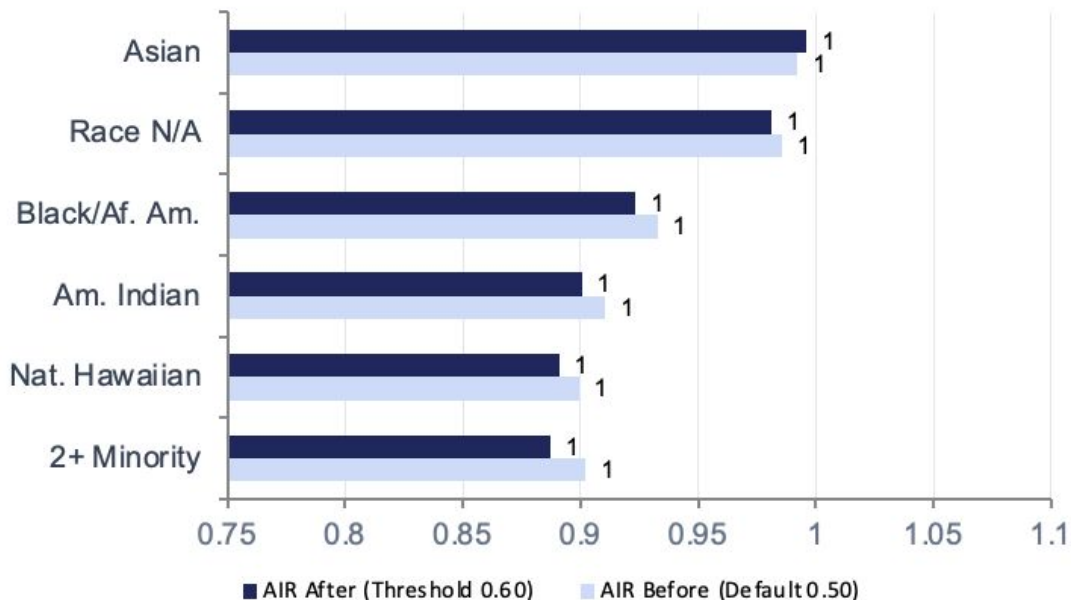
MITIGATION

- Mitigation Strategy: Threshold Adjustment
- Threshold Optimization Results
- Fairness Before vs After Mitigation
- Performance–Fairness Tradeoffs



Mitigation — Before vs After

Threshold adjustment from 0.50 → 0.60 · Transparent, auditable, no retraining required



0.60

Optimal Threshold

Selected from 0.30–0.60 sweep

0.7233

Macro F1 After

Improved from 0.7012 baseline

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Groups Pass AIR

All above 0.80 threshold

Fairness and performance improved together — threshold optimization raised Macro F1 while maintaining all AIR thresholds.

AGENDA - PHASE 5

ROBUSTNESS - PHASE 5

- Stress Testing Scenarios
- Model Performance Under Stress

FINALLY:

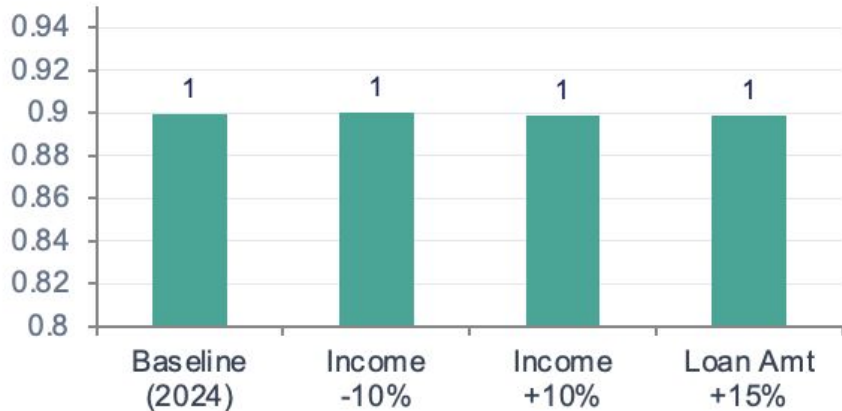
- Residual Risks & Limitations
- Final Conclusions & Key Takeaway



Risks That Remain After Mitigation

Documented uncertainties — not incomplete work

Min AIR — 4 Stress Scenarios (2025 Simulation)



All 4 scenarios: LOW RISK • Min AIR stays above 0.89 under every simulated shift

Residual Risk Register

- HIGH** High FPR (50–70% all groups)
Monitor approval rates monthly
- HIGH** Proxy leakage via income/property
SHAP quarterly monitoring
- HIGH** 2025 distribution shift
Documented — beyond simulation
- MED** Thin subgroups (N < 500)
Flag in reporting, not decision
- LOW** interest_rate removed
Accepted — fairness tradeoff

Stress Testing



KEY RESULT - All scenarios → LOW risk

Is Deployment Defensible?

CONDITIONAL APPROVAL FOR DEPLOYMENT

- | | |
|---|---|
| ✓ Optimization objective documented (Macro F1 + AIR ≥ 0.80) | ✓ Intersectional analysis (Race $\times$ Sex) completed |
| ✓ Baseline model established (Random Forest) | ✓ Threshold mitigation applied (0.60) |
| ✓ Class imbalance addressed (scale_pos_weight = 3.12) | ✓ 4 stress scenarios tested — all LOW risk |
| ✓ All 8 race groups pass AIR ≥ 0.80 | ✓ Residual risks documented |
| ✓ FPR / FNR measured by race — all groups | ✓ Governance & monitoring plan established |

MONITORING TRIGGERS: AIR < 0.80 $\rightarrow$ suspend within 5 days | PSI > 0.20 $\rightarrow$ retrain | F1 < 0.65 $\rightarrow$ full review | **Owner: Model Risk Committee**

Residual Risk

Risk	Severity Mitigation Applied	Status
Remaining AIR gaps near threshold	Medium Yes	Monitor monthly AIR
Proxy leakage (income, property value)	High Partial	SHAP proxy monitoring
Interest rate removed -reduces model signal	Low Yes	Accepted trade off for fairness
2025 distribution shift (unquantified)	High No	Documented- cannot quantify yet
Thin group estimates	Medium No	Flag in reporting
Intersectional gaps not fully resolved	Medium Partial	Requires more data collection

Key Takeaway: proxy bias, distribution shift risk, and small subgroup instability still exist, the model is considered conditionally deployable because it meets the AIR threshold, the risks are documented, and a monitoring plan is in place.



TO CONCLUDE:

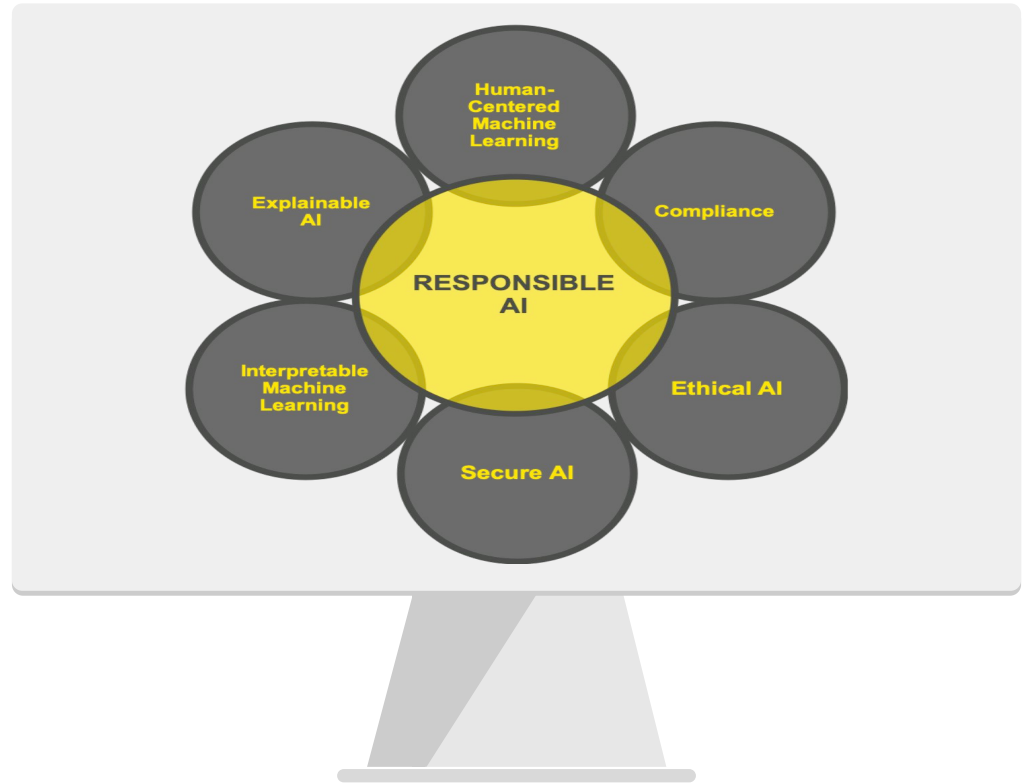
A good model is not one without risk — it is one where risks are measured, mitigated, and documented.





Thank you!

ANY QUESTIONS?



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Optimization Objective

What Is The Optimization Objective ?

- Predict loan approval (approved vs denied)
- Optimize:
 - ROC-AUC (0.86)
 - Macro F1 (0.79 balanced performance)
- Balance:
 - Accuracy **AND** fairness

This project focuses on auditing a loan approval model using HMDA data. We evaluate not just performance, but also fairness, robustness, and potential risks using methods from the course.

***We built a loan approval model → found fairness risks → fixed them
→ tested if they hold under stress → identified what risks remain***

Fairness Metrics

AIR < 0.80 = legal concern

- AIR (Adverse Impact Ratio)
- FPR
- FNR

AIR by Race

All groups pass AIR ≥ 0.80 Threshold

Generating AIR comparison plot...



Saved: air_by_race.png

RESPONSIBLE MACHINE LEARNING CAPSTONE PROJECT

MITIGATION - PHASE 4

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- Fairness Before vs After Mitigation
- Performance–Fairness Tradeoffs



RESPONSIBLE MACHINE LEARNING CAPSTONE PROJECT

ROBUSTNESS - PHASE 5

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- Model Performance Under Stress

FINALLY:

- Residual Risks & Limitations
- Final Conclusions & Key Takeaway





Mitigation Strategy

Adjusted decision threshold

- No retraining needed
- Improves fairness directly

Threshold Selection

... Testing thresholds from 0.30 to 0.60...

Threshold Analysis:

Threshold	Groups Failing	AIR	Macro F1
0.30	0	0	0.6535
0.32	0	0	0.6593
0.34	0	0	0.6653
0.36	0	0	0.6712
0.38	0	0	0.6760
0.40	0	0	0.6802
0.42	0	0	0.6844
0.44	0	0	0.6886
0.46	0	0	0.6929
0.48	0	0	0.6972
0.50	0	0	0.7012
0.52	0	0	0.7048
0.54	0	0	0.7085
0.56	0	0	0.7136
0.58	0	0	0.7183
0.60	0	0	0.7233

Best threshold selected: 0.6

Groups failing AIR: 0.0

Macro F1: 0.7233

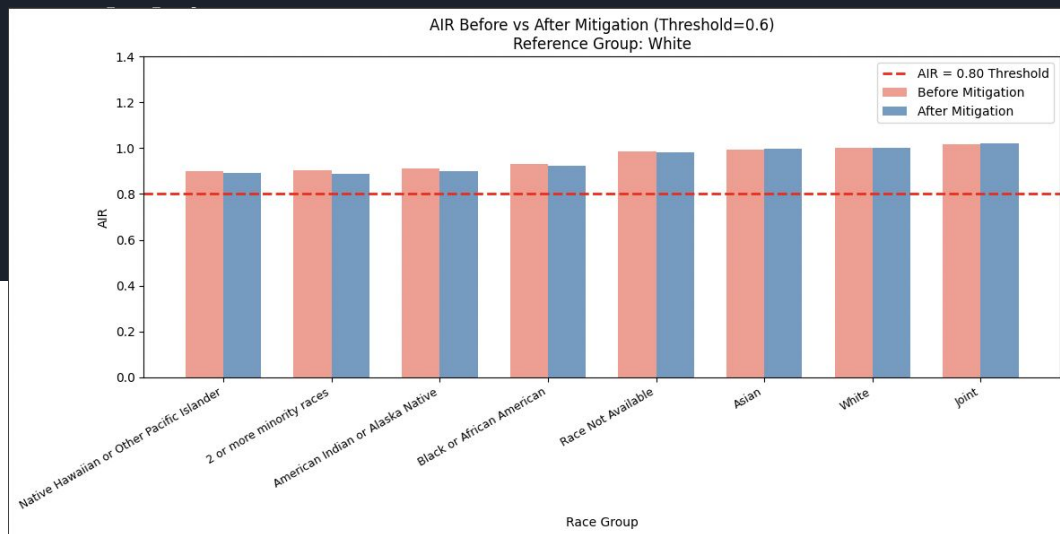
Best threshold = 0.60

Before vs After

All groups pass fairness
after mitigation

Small tradeoff in FNR

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RESPONSIBLE MACHINE LEARNING CAPSTONE PROJECT

ROBUSTNESS - PHASE 5

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